

Proposed Prompt for Antigravity Agent

Role: Expert Software Architect specializing in the ICTM Framework.

Objective: Develop a comprehensive technical development plan and code execution strategy to develop CMSys using ICTM Framework.

Project Context: CMSys is an ICTM Framework-based CMS providing WordPress-like functionality. It utilizes a modular, hook-driven architecture, ensuring feature expansion does not necessitate core framework modifications.

1. Architectural & Development Standards

- **PHP Syntax Compliance:** All code must strictly use standard brace-based control structures (e.g., `if (...) { ... }`, `foreach (...) { ... }`). The use of alternative template syntax (e.g., `if(): ... endif;`) is strictly forbidden.
- **Event-Driven Core:** Implement a robust `HookManager`. All modules and themes must interface with core logic exclusively via `do_action()` and `apply_filters()` to maintain core file integrity.
- **Security Architecture:**
 - **Password Hashing:** Use `password_hash()` with `PASSWORD_DEFAULT` algorithm.
 - **Authentication:** Must utilize existing `ICTM_Auth` session management.

2. Module & Theme Lifecycle Management

- **Manifest Specification:** Every entity must contain a `manifest.json` defining: name, version, author, dependencies, and required SQL scripts.
- **Installation/Uninstallation Engine:**
 - Validate ZIP structure upon upload.
 - Extract to `/Themes/` or `/Modules/` directories.
 - Execution of `schema.sql` and global registry updates.
 - Uninstallation must include a complete cleanup of module-specific DB tables and file directories.
- **Atomic Constraints:**
 - Support dual theme storage (`frontend_theme/backend_theme`) in `site_options`.
 - Atomic activation: One theme active per scope; activating a new theme must trigger immediate deactivation of the predecessor.

3. Content & Media Management

- **Editor Interface:** Integrate Summernote WYSIWYG editor with media insertion capabilities.

- **Post Lifecycle:** Support Draft, Published, Expired, and Scheduled statuses.
- **Slug Generation:** Implement automated sanitization (lowercase, special character removal) with collision checks and manual override fields.
- **Media Library System:**
 - **Storage:** Files must reside at `Writables/images/YYYY/MM/DD/`.
 - **DB Integrity:** Metadata (original name, path, mime-type, dimensions, user_id) must be logged in a `media_library` table.
 - **Prohibition:** No binary/blob storage or base64-encoded strings allowed for media.
 - **Cleanup:** Deletion of any post/page must trigger an automatic recursive cleanup of associated physical media files and DB records.

4. RBAC & User Management

- **Admin:** Full read/write/configuration access.
- **Editor:** Access to Content, Media, and Comments. Restricted from System Settings or Theme/Module installation.
- **Subscriber:** Frontend profile access only. No access to `/admin/*`.

5. System Dashboard & Settings

- **Settings Engine:** CRUD operations via a centralized `site_options` (Key-Value) table.
- **SEO & Analytics:** Provide fields for `<meta>` tag injection and footer/body script insertion.
- **Dashboard Aggregation:** Single controller for aggregated statistics (posts, pages, users, comments), utilizing caching to optimize high-traffic performance.

6. Frontend & Admin UI/UX

- **Framework:** Strict adherence to Bootstrap 5.3 grid and component systems.
- **Data Handling:** Admin index views must implement DataTables with server-side processing for pagination and search.
- **Interactivity:**
 - Status toggles (e.g., Publish/Draft) must utilize asynchronous Ajax calls to `/admin/api/update-status` for real-time UI updates without reloads.
 - Integrate Tagify for taxonomy/keyword inputs.
 - Use Wow-animate on DOM load for aesthetic effects.
- **Dashboard:** Aggregated statistics controller with caching.
- **Frontend Framework:** Strict adherence to Bootstrap 5.3.
- **Interactivity:** Admin index views must use DataTables with server-side processing; status toggles must use Ajax calls; include Tagify for taxonomy and Wow-animate for aesthetic effects.